

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456728.74166 +/- 0.00073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1 +/- 0.045
Transit depth (Rp/Rs)^2	1.0 +/- 0.9 [%]
Orbital Inclination (inc)	77.5 +/- 2.94
Airmass coefficient 1 (a1)	1.016 +/- 0.047
Airmass coefficient 2 (a2)	-0.015 +/- 0.082
Scatter in the residuals of the lightcurve fit is	4.56 %
Best Comparison Star	None
Optimal Aperture	10.85
Optimal Annulus	26.21
Transit Duration (day)	0.02 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1 ± 0.045 vs 0.1594 (deviation 0.92σ)
Duration fit vs published	0.02 d vs 0.0512 d (deviation 1.15σ)
Mid-time O-C	-1.9 min
Depth SNR	1.6
Residual scatter	4.56 %

Light curve

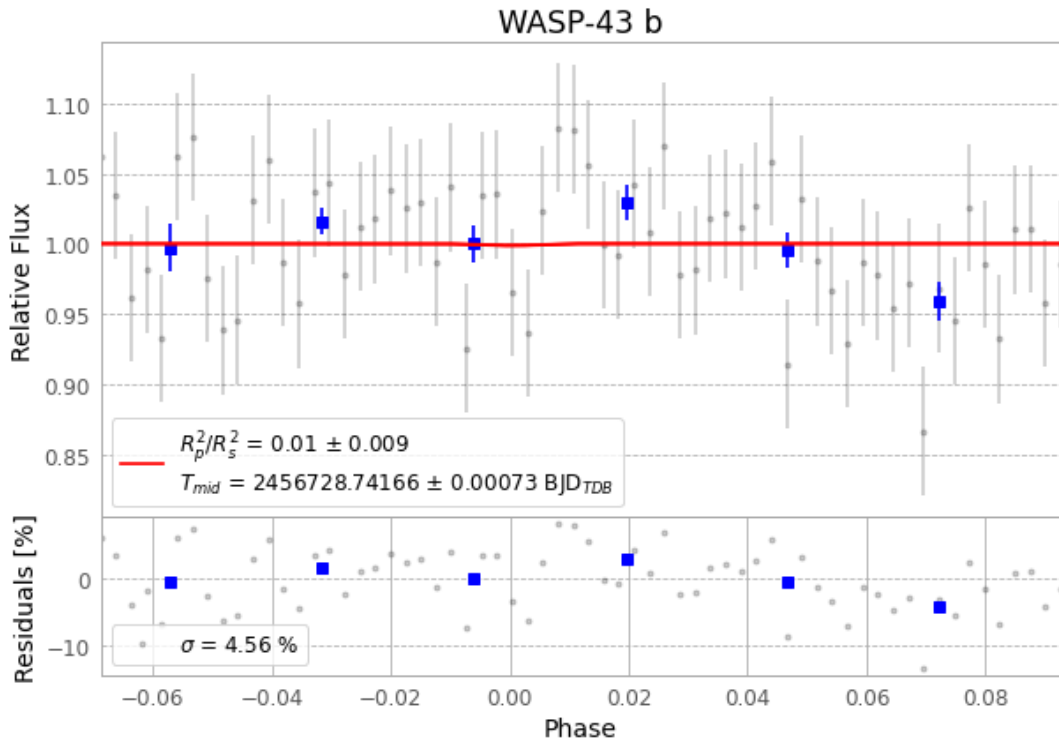


Figure 1 — FinalLightCurve_WASP-43 b_2014-03-11.png (SHA-256 294fdf5835a7bf6a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 332], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7e174244ccbc9124...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140312041812.FITS → WASP-43140312072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140312.log (f745a08b37b8...), FinalLightCurve_WASP-43 b_2014-03-11.png (294fdf5835a7...), FinalLightCurve_WASP-43 b_2014-03-11.csv (414aa3723ec6...)

Compute resources

Wall time 128.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:38Z.